



Department of Computer Science & Engineering

LAB MANUAL

22CS025 - Algorithm Design & Implementation-5Sem-2022

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Course Name	22CS025 - Algorithm Design & Implementation-5Sem-2022



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Aim: Matrix Chain Multiplication problem

A matrix can be represented as a 2-dimensional array. Two matrices are eligible for multiplication if the number of columns in 1st matrix is equal to the number of rows in 2nd matrix.

The total number of operations required for multiplying two matrices A [m x n] and B [n x o] will be $(m * n * o)$.

We have many ways to multiply a chain of matrices because matrix multiplication is associative, each way require different number of operations to complete. Mr. Amit has a sequence of matrices and interested to find the way in which minimum operations are required to multiply all matrices. The problem is not actually to perform the multiplications, but merely to decide in which order to perform the multiplications.

Example :

If we had four matrices A, B, C, and D, we would have:

(ABC)D or (AB)(CD) or A(BCD) and we can use the same kind of parentheses for inner problems to get ((A(BC))D) or (((AB)C)D) or (AB)(CD) or (A((BC)D)) or (A(B(CD)))

Suppose A is a 10×20 matrix, B is a 20×30 matrix, and C is a 30×40 matrix. Then,

For 3 matrices we have 2 options

(A(BC)) = To Multiply BC we need $20*30*40=24000$ and we get 20×40 matrix, then it is multiplied with A and take $10*20*40=8000$ operations, so total 32000 operations needed.

((AB)C) = To Multiply AB we need $10*20*30=6000$ and we get 10×30 matrix, then it is multiplied with C and take $10*30*40=12000$ operations, so total 18000 operations needed.

Input Format:

First Line will contain an integer N denoting the number of matrices.

Second line contains N+1 integers denoting the matrix sizes separated by space.

Output Format:

Print the minimum operations needed for matrix multiplication as specified above.

Constraints

$2 \leq N \leq 100$

$1 \leq \text{arr}[i] \leq 500$

Sample Input

3

10 20 30 40

Sample Output

18000

Solution:

```
class Result
{
    // Matrix A[i] has dimension p[i-1] x p[i] for i = 1..n
    static int matrixChainMultiplication(int p[], int n){
        // Write your code here
        Integer[][] dp = new Integer[n+1][n+1];
        for(int i =0;i<=n;i++){
            for(int j=0;j<=n;j++){
                if(i!=j){
                    dp[i][j]=-1;
                }
                if(i==j){
                    dp[i][j]=0;
                }
            }
        }
    }
}
```

```
    }  
  }  
}  
return ans(p,1,n,dp);  
}  
static int ans(int p[],int i,int j,Integer[][] dp){  
  if(i==j){  
    return 0;  
  }  
  if(dp[i][j]!=-1){  
    return dp[i][j];  
  }  
  int r=Integer.MAX_VALUE;  
  for(int k=i;k<j;k++){  
    int a = ans(p,i,k,dp);  
    int b = ans(p,k+1,j,dp);  
    int c = p[i-1]*p[k]*p[j];  
    r=Math.min(r,a+b+c);  
  }  
  dp[i][j]=r;  
  return dp[i][j];  
}  
}
```



CodeQuotient